Adversarial Attacks on Text-Dependent Speaker Verification: A Comparative Study Using WavLM and ECAPA

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Abstract

Adversarial attacks on text-independent speaker verification (TI-SV) often assume access to the genuine speaker's enrollment utterance, which is impractical. Text-dependent speaker verification (TD-SV) adds the constraint of a fixed password, making attacks more realistic and challenging. This work studies embedding-level adversarial attacks, where perturbations are applied to non-password speech to bypass TD-SV systems. Embedding-level attacks are computationally efficient and interpretable, allowing analysis of the embedding space without modifying raw audio directly. We evaluate ECAPA-TDNN and WavLM embeddings under a white-box FGSM attack on the TIMIT dataset, quantifying performance using Equal Error Rate (EER) and Attack Success Rate (ASR). Results show that WavLM is highly sensitive, while ECAPA-TDNN maintains robustness, achieving an ASR of up to 64.28%. These findings highlight critical vulnerabilities in TD-SV systems and emphasize the need for more robust defenses..

Keywords: Text-dependent speaker verification, adversarial attacks, ECAPA-TDNN, WavLM

1 Introduction

Speaker verification (SV) determines whether a speech utterance matches a claimed speaker and is widely deployed in security-sensitive applications such as voice-based authentication and access control. Text-dependent speaker verification (TD-SV) constrains the lexical content of the utterance, enabling tighter phonetic alignment and improved discrimination compared to text-independent systems. This approach has gained renewed attention through recent benchmark challenges and transformer-based architectures. Ensuring robustness is therefore critical for real-world TD-SV deployments.

Modern TD-SV systems rely on discriminative neural speaker embeddings. ECAPA-TDNN enhances speaker modeling through emphasized channel attention and multi-scale temporal aggregation, while self-supervised models such as WavLM leverage large-scale pretraining to generalize across diverse speech tasks. Despite their strong performance, these systems remain vulnerable to adversarial attacks, where imperceptible perturbations can significantly degrade verification accuracy. Most existing studies focus on waveform-level attacks or text-independent SV, leaving embedding-level adversarial robustness in TD-SV relatively underexplored.

In this work, we investigate embedding-level adversarial attacks on TD-SV systems, performing a comparative study of ECAPA-TDNN and WavLM embeddings under a white-box Fast Gradient Sign Method (FGSM) threat model. Using fixed-phrase TIMIT SA utterances, we evaluate system robustness in terms of Equal Error Rate (EER) and Attack Success Rate (ASR). Our results show that ECAPA-TDNN embeddings exhibit substantially higher robustness than WavLM under the same attack conditions.

The main contributions of this work are:

- A systematic evaluation of embedding-level adversarial robustness in TD-SV.
- A comparative analysis of ECAPA-TDNN and WavLM embeddings under FGSM attacks.
- An analysis of cosine similarity score distributions to explain observed differences in robustness.

2 Related Work

2.1 Text-Dependent Speaker Verification

Text-dependent speaker verification aims to confirm a speaker's identity under a fixed lexical constraint, improving phonetic alignment and discrimination compared to text-independent approaches. Early TD-SV systems relied on generative models such as Gaussian mixture models and hidden Markov models, but recent advances favor deep neural networks that learn speaker-discriminative representations directly from speech. Convolutional and time-delay neural network architectures, particularly ECAPA-TDNN, have shown strong performance by modeling temporal context effectively and enhancing embedding discriminability.

2.2 Self-Supervised Speech Representations

Self-supervised learning has become an effective paradigm for speech representation, enabling models to leverage large-scale unlabeled data. Models like WavLM capture contextual, phonetic, and speaker information, and transfer well to downstream tasks including speaker verification. Compared to supervised embeddings, self-supervised representations provide improved generalization across domains. However, their robustness to adversarial perturbations in TD-SV remains underexplored.

2.3 Adversarial Attacks on Speaker Verification

Adversarial attacks, initially studied in image classification, have been adapted to speech and speaker verification. Carefully crafted perturbations can significantly degrade verification performance. Most prior work focuses on text-independent systems or evaluates a single architecture, limiting applicability to TD-SV. Waveform-level attacks are often computationally expensive and sensitive to preprocessing. Embedding-level attacks offer an efficient and interpretable alternative, allowing fair comparisons across different embeddings under controlled conditions. Differences in embedding geometry—where supervised models prioritize discriminative speaker boundaries and self-supervised models encode broader contextual information—can influence vulnerability to perturbations.

2.4 Motivation and Contribution

Despite progress in adversarial robustness for speaker verification, systematic evaluation under text-dependent constraints is limited. This work addresses this gap by comparing embedding-level attacks on ECAPA-TDNN and WavLM embeddings, isolating the effects of representation paradigm and architecture on robustness. Our analysis highlights how embedding geometry and discriminability influence vulnerability in TD-SV systems.

- **WavLM** is a transformer-based self-supervised model capturing rich phonetic and speaker information through masked speech prediction and denoising objectives, improving downstream speaker verification performance.
- **ECAPA-TDNN** is a convolutional TDNN optimized for speaker verification with multi-scale feature aggregation and channel attention, enhancing embedding discriminability and robustness.

3 Methodology

3.1 Text-Dependent Speaker Verification Framework

We consider a **text-dependent speaker verification** (TD-SV) scenario, in which both enrollment and verification utterances correspond to a **fixed lexical content**, enabling strict phonetic alignment between utterances and reducing linguistic variability [15, 16]. Such a setting is commonly adopted in TD-SV benchmarks and challenge evaluations, where predefined passphrases or fixed phrases are used to improve verification accuracy under short-utterance conditions and to constrain the phonetic variability inherent to speech signals [16].

All experiments are conducted using the **SA utterances** from the TIMIT corpus, which consist of predefined sentences that are identical across all speakers and are widely used for controlled evaluation of text-dependent speaker verification systems.

Given an enrollment utterance $\mathbf{x}_e$ and a test utterance $\mathbf{x}_t$, the objective of TD-SV is to determine whether both utterances were produced by the same speaker under the constraint that $\mathbf{x}_e$ and $\mathbf{x}_t$ share identical lexical content.

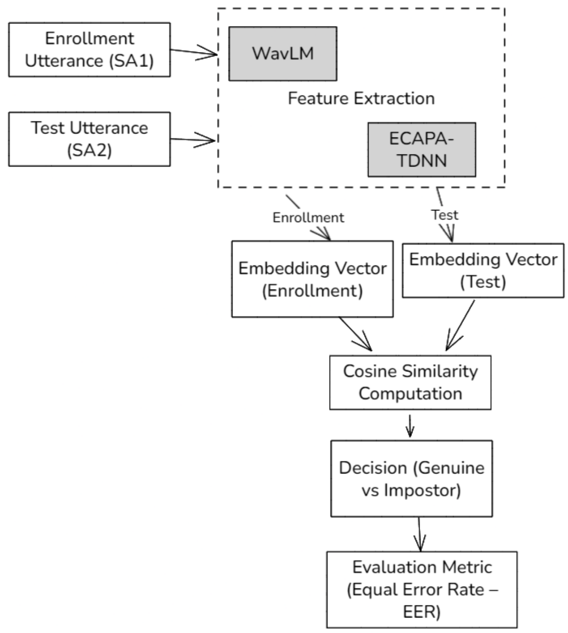

Figure 1: TD-SV pipeline. Enrollment and test utterances are embedded and compared using cosine similarity.

3.2 Dataset and Protocol

For each speaker, two utterances of the same phrase are used:

- **SA1** as the enrollment utterance.
- **SA2** as the test utterance.

This ensures a strict text-dependent condition, a clear separation between enrollment and test data, and a realistic evaluation of speaker discrimination. Enrollment–test pairs generate **genuine trials** (same speaker) and **impostor trials** (different speakers), isolating speaker modeling performance.

3.3 Speaker Embedding Extraction

Speaker embeddings are extracted using pre-trained models. We evaluate:

- **WavLM** [1], a transformer-based self-supervised model capturing rich contextual, phonetic, and speaker information through masked speech prediction and denoising objectives. WavLM has been shown to improve robustness and speaker representation quality in downstream tasks including text-dependent speaker verification.

- **ECAPA-TDNN** [2], a convolutional TDNN optimized for speaker verification with multi-scale feature aggregation and channel attention. This architecture enhances embedding discriminability and robustness, outperforming standard TDNNs in TD-SV benchmarks.

For each utterance, a fixed-dimensional embedding is L2-normalized. Verification uses the cosine similarity between enrollment $\mathbf{e}$ and test embedding $\mathbf{t}$:

$$s(\mathbf{e}, \mathbf{t}) = \frac{\mathbf{e}^\top \mathbf{t}}{\|\mathbf{e}\|_2 \|\mathbf{t}\|_2}. \tag{1}$$

A decision is made using threshold τ:

$$\mathcal{D}(s) = \begin{cases} \text{accept}, & s \geq \tau, \\ \text{reject}, & s < \tau. \end{cases} \tag{2}$$

3.4 Adversarial Threat Model

We consider a **white-box, targeted embedding-level attack** [11, 12], where the attacker aims to degrade verification for a specific target speaker. The attacker has access to the test embedding and the target enrollment embedding.

Formally, the adversary seeks a perturbation δ such that:

$$\mathcal{D}(s(\mathbf{e}_t, \mathbf{t} + \delta)) \neq \mathcal{D}(s(\mathbf{e}_t, \mathbf{t})). \tag{3}$$

3.5 Adversarial Attack Method

We generate adversarial embeddings using the **Fast Gradient Sign Method (FGSM)** [11]:

$$\mathbf{t}' = \mathbf{t} + \epsilon \cdot \text{sign}\left(\nabla_{\mathbf{t}} \mathcal{L}(\mathbf{e}, \mathbf{t})\right), \quad \mathcal{L}(\mathbf{e}, \mathbf{t}) = -s(\mathbf{e}, \mathbf{t}). \tag{4}$$

The perturbation magnitude ϵ controls attack strength. No adversarial noise is added at the waveform level, and model parameters remain fixed. This method is applied identically to WavLM and ECAPA embeddings.

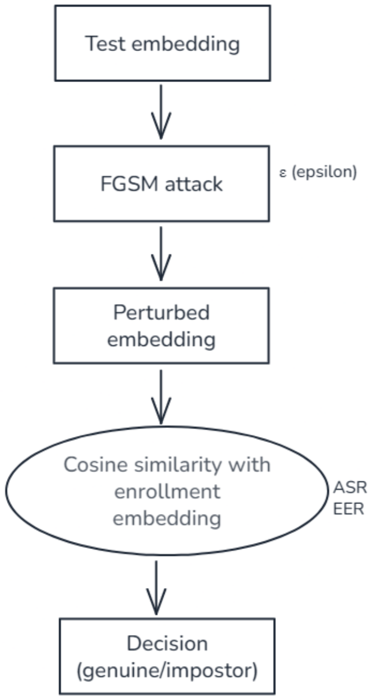

Figure 2: Embedding-level adversarial perturbation using FGSM.

3.6 Robustness Evaluation Metrics

We assess adversarial robustness using:

Equal Error Rate (EER): Operating point where false acceptance rate equals false rejection rate:

$$\text{EER} = \text{FAR}(\tau^)) = \text{FRR}(\tau^)), \tag{5}$$

Attack Success Rate (ASR): Proportion of trials whose verification decision changes due to the attack:

$$\text{ASR} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[\mathcal{D}(s_i) \neq \mathcal{D}(s_i')\right]. \tag{6}$$

By evaluating EER and ASR across varying ϵ, we quantify baseline discriminative power, embedding-specific vulnerability, and robustness differences between WavLM and ECAPA.

4 Experiments

4.1 Overview

We evaluate the robustness of text-dependent speaker verification (TD-SV) systems using WavLM [1] and ECAPA-TDNN [2] embeddings under embedding-level FGSM attacks [11, 12].

System performance is quantified using **Equal Error Rate (EER)** and **Attack Success Rate (ASR)** across varying attack strengths ϵ.

4.2 Dataset and Data Structure

Experiments are conducted on the TIMIT corpus [14], using only the SA utterances under a strict text-dependent protocol. This setup ensures a fixed lexical content for enrollment and verification utterances, allowing precise phonetic alignment and minimizing linguistic variability, which is essential for evaluating targeted embedding-level attacks. For each speaker:

- **SA1** is used as the enrollment utterance.
- **SA2** is used as the test utterance.

Enrollment and test utterances are paired to generate trials:

- **Genuine trials:** Same speaker, identical phrase (SA1 vs SA2).
- **Impostor trials:** Different speakers, identical phrase.

This setup isolates speaker-specific characteristics while maintaining strict text-dependent conditions. The dataset contains N speakers and M total trials, with a balanced distribution of genuine and impostor trials.

4.3 Metrics and Evaluation Protocol

System performance is evaluated using:

- **Equal Error Rate (EER):** Operating point where false acceptance rate equals false rejection rate.
- **Attack Success Rate (ASR):** Proportion of trials whose verification decision changes after the attack.

EER and ASR are computed for both WavLM and ECAPA-TDNN embeddings across varying ϵ values.

4.4 Baseline Verification Performance

Baseline system performance (no adversarial perturbation) is summarized in Table 1.

Table 1: Baseline EER (%) for WavLM and ECAPA-TDNN embeddings.

Embedding	EER (%)
WavLM	0.1375
ECAPA-TDNN	0.0042

4.5 Adversarial Attack Results

Figures 3 and 4 illustrate the impact of embedding-level FGSM attacks on TD-SV performance. EER (%) and ASR (%) are plotted versus attack strength ϵ for ECAPA-TDNN and WavLM embeddings, respectively.

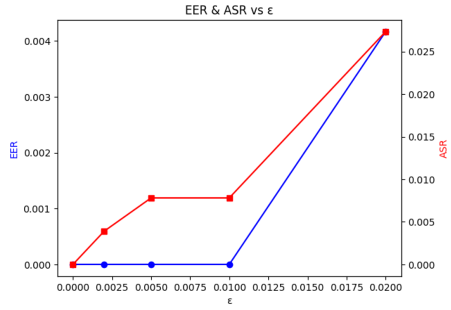

Figure 3: ECAPA-TDNN performance under embedding-level FGSM attacks. Robust verification is maintained for small perturbations.

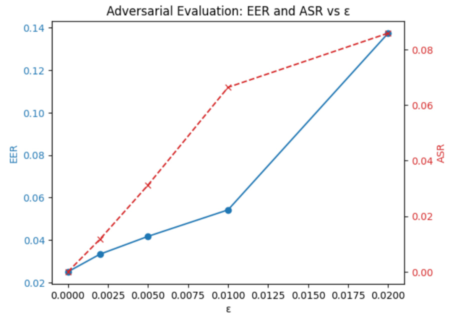

Figure 4: WavLM performance under embedding-level FGSM attacks. Verification performance degrades faster with increasing ϵ.

Numeric results are summarized in Table 4.5, showing that ECAPA-TDNN maintains stable performance under small to moderate attacks, while WavLM EER and ASR increase more rapidly.

[t]

ϵ	ECAPA-TDNN		WavLM	
	EER	ASR	EER	ASR
0.000	0.00	0.00	2.50	0.00
0.002	0.00	0.39	3.33	1.17
0.005	0.00	0.78	4.17	3.12
0.010	0.00	0.78	5.42	6.64
0.020	0.42	2.73	13.75	8.59

4.6 Score Distribution Analysis

We analyze cosine similarity score distributions to explain robustness differences.

For WavLM embeddings, baseline scores show clear separation between genuine and impostor trials. Under $\epsilon = 0.01$, genuine scores decrease significantly while impostor scores remain stable, reducing score margin and increasing EER and ASR (Table 2).

Table 2: WavLM cosine similarity score statistics before and after adversarial attack ($\epsilon = 0.01$).

	Genuine	Impostor
Baseline mean	0.946	0.622
Baseline std	0.020	0.191
Adversarial mean	0.882	0.584
Adversarial std	0.027	0.185

ECAPA-TDNN embeddings exhibit a larger separation and lower variance, explaining higher robustness under attacks (Table 3).

Table 3: ECAPA-TDNN cosine similarity score statistics under baseline conditions.

Class	Mean	Std
Genuine	0.667	0.079
Impostor	0.108	0.085

4.7 Observations

- ECAPA-TDNN embeddings demonstrate higher robustness due to larger inter-class margins and lower intra-class variance.

- WavLM embeddings are more sensitive to FGSM perturbations, resulting in faster increases in EER and ASR.

- Score distribution analysis confirms that embedding architecture and discriminability strongly influence adversarial robustness.

5 Discussion

The experimental results reveal a clear contrast in adversarial robustness between ECAPA-TDNN and WavLM embeddings in text-dependent speaker verification (TD-SV). While both models achieve competitive baseline performance, their behavior under embedding-level FGSM attacks differs significantly.

WavLM embeddings show a rapid increase in both EER and ASR as the perturbation magnitude ϵ grows. This sensitivity can be attributed to the nature of self-supervised representations, which prioritize broad phonetic and contextual information rather than explicit speaker discrimination. As a result, small perturbations are sufficient to reduce the similarity margin between genuine and impostor trials, leading to verification errors.

In contrast, ECAPA-TDNN demonstrates strong robustness for small and moderate attack strengths. Score distribution analysis indicates that ECAPA-TDNN embeddings maintain a larger separation between genuine and impostor classes, with lower intra-class variance. This wider margin provides inherent resistance to small adversarial perturbations, explaining the near-zero EER observed for low ϵ values.

These findings suggest that embedding geometry plays a critical role in adversarial robustness. Architectures explicitly optimized for speaker discrimination, such as ECAPA-TDNN, appear more resilient to embedding-level attacks than general-purpose self-supervised models. This observation has important implications for the deployment of TD-SV systems in security-sensitive applications.

6 Limitations and Future Work

Despite the insights obtained in this study, several limitations should be noted.

First, adversarial perturbations are generated at the **embedding level** rather than on the raw waveform. While this enables a controlled, model-agnostic analysis of representation robustness, it does not fully reflect real-world attack scenarios. Future work will extend this study to **waveform-level attacks**, providing a more realistic assessment of system vulnerability.

Second, the evaluation uses a **strict text-dependent protocol** with only SA utterances from the TIMIT corpus. Although this isolates speaker-specific characteristics, it limits generalization to other phrases and acoustic conditions. Moreover, **publicly available datasets for text-dependent adversarial speaker verification are scarce**. Future research will focus on **creating dedicated TD-SV datasets** to enable more comprehensive evaluations.

Third, the current threat model assumes a **white-box adversary** with access to speaker embeddings. While this represents a worst-case scenario, real-world attackers may operate under partial or no knowledge. Exploring **black-box, transfer-based, and limited-knowledge attacks** will provide a more complete understanding of practical risks.

Additionally, only the **Fast Gradient Sign Method (FGSM)** is considered. Although FGSM is computationally efficient, more advanced iterative or optimization-based attacks could further challenge system robustness. Future work will investigate **iterative adversarial attacks** and **defense strategies**, including adversarial training and robust score normalization.

Finally, this study considers only two embedding models. Extending the analysis to additional **recent speaker representation architectures** would further elucidate the relationship between embedding design, speaker discriminability, and adversarial robustness.

7 Conclusion

This work presented a systematic analysis of embedding-level adversarial robustness in text-dependent speaker verification systems. We evaluated ECAPA-TDNN and WavLM embeddings under FGSM attacks using a strict TD-SV protocol on the TIMIT dataset.

Experimental results indicate that WavLM embeddings are highly sensitive to adversarial perturbations, exhibiting rapid degradation in verification performance as attack strength increases. In contrast, ECAPA-TDNN maintains strong robustness, with near-zero EER and low ASR for small to moderate perturbation magnitudes.

Analysis of cosine similarity distributions shows that ECAPA-TDNN produces embeddings with larger inter-class separation and lower intra-class variance, resulting in a stronger decision margin. This embedding geometry renders small adversarial perturbations insufficient to alter verification outcomes, highlighting the importance of architecture and training paradigm for securing TD-SV systems.

Future work will extend this study to waveform-level attacks, black-box threat models, and alternative self-supervised representations. Incorporating adversarial training and other defense mechanisms remains a promising direction for enhancing the security of TD-SV systems in real-world applications.

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[0]**Code availability:** `https://github.com/oumas3/TDSV_SYSTEMS_ECAPA_WAVLM`